

Activity: Find files with Linux commands

Objective of the Lab

Practice using Linux Bash commands to:

- Navigate directories
- Locate files
- Read file contents
- Analyze information inside files

These skills are important for security analysts, especially when:

- Investigating unauthorized access
- Checking user activity
- Reading system logs
- Working on remote systems without GUI

Scenario

In this scenario, you have to locate and analyze the information of certain files located in the /home/analyst directory.

Here's how you'll do this: **First**, you'll get the information of the current working directory you're in and display the contents of the directory. **Second**, you'll navigate to the reports directory and list the subdirectories it contains. **Third**, you'll navigate to the users subdirectory and display the contents of the Q1_added_users.txt file. **Finally**, you'll navigate to the logs directory and display the first 10 lines of a file it contains.

Task 1. Get the current directory information

1. Display your working directory

```
analyst@ed1f90223c80:~$ pwd
/home/analyst
analyst@ed1f90223c80:~$
```

2. Display the names of the files and directories in the current working directory.

```
analyst@ed1f90223c80:~$ ls
logs  projects  reports  temp
analyst@ed1f90223c80:~$
```

How many directories does the current working directory contain?

- ☐ Five
- ☒ Four
- ☐ One
- ☐ Two

Submit

Which directory is your current working directory?

- ☒ /home/analyst
- ☐ /var/logs
- ☐ /home/analyst/logs
- ☐ /home

Submit

Task 2. Change directory and list the subdirectories

1. Navigate to the /home/analyst/reports directory.

```
analyst@ed1f90223c80:~$ cd /home/analyst/reports
analyst@ed1f90223c80:~/reports$
```

2. Display the files and subdirectories in the /home/analyst/reports directory.

```
analyst@ed1f90223c80:~/reports$ ls
users
analyst@ed1f90223c80:~/reports$
```



What is the name of the subdirectory in the /home/analyst/reports directory?

☐ logs

☐ projects

☒ users

☐ analyst

Submit

Task 3. Locate and read the contents of a file

1. Navigate to the /home/analyst/reports/users directory.

```
analyst@ed1f90223c80:~/reports$ cd /home/analyst/reports/users
analyst@ed1f90223c80:~/reports/users$
```

2. List the files in the current directory.

```
analyst@ed1f90223c80:~/reports/users$ ls
Q1_added_users.txt  Q1_deleted_users.txt
analyst@ed1f90223c80:~/reports/users$
```

3. Display the contents of the Q1_added_users.txt file.

```
analyst@ed1f90223c80:~/reports/users$ cat Q1_added_users.txt
employee_id  username  department
1001         bmoreno  Marketing
1026         apatel   Human Resources
1041         cgriffin Sales
1104         mreed    Information Technology
1177         aezra     Human Resources
1188         noshiro   Finance
analyst@ed1f90223c80:~/reports/users$
```



What is the employee_id of the user mreed in the Information Technology department?

 1104

☐ 1001

☐ 1177

☐ 1188

Submit



What department does the employee with the username aezra work in?

 Human Resources

☐ Information Technology

☐ Finance

☐ Sales

Submit

Task 4. Navigate to a directory and locate a file

1. Navigate to the /home/analyst/logs directory.

```
analyst@ed1f90223c80:~/reports/users$ cd /home/analyst/logs
analyst@ed1f90223c80:~/logs$
```

2. Display the name of the file it contains.

```
analyst@ed1f90223c80:~/logs$ ls
server_logs.txt
analyst@ed1f90223c80:~/logs$
```

3. Display the first **10** lines of this file.

```
analyst@ed1f90223c80:~/logs$ head server_logs.txt
2022-09-28 13:55:55 info    User logged on successfully
2022-09-28 13:56:22 error   The password is incorrect
2022-09-28 13:56:48 warning The file storage is 75% full
2022-09-28 15:55:55 info    User logged on successfully
2022-09-28 15:56:22 error   The username is incorrect
2022-09-28 15:56:48 warning The file storage is 90% full
2022-09-28 16:55:55 info    User navigated to settings page
2022-09-28 16:56:22 error   The password is incorrect
2022-09-28 16:56:48 warning The current user's password expires in 15 days
2022-09-29 13:55:55 info    User logged on successfully
analyst@ed1f90223c80:~/logs$
```

How many warning messages are in the first 10 lines of the server_logs.txt file?

☐ One

☐ Two

☐ Six

☒ Three

Submit

FULL VIEW

```
analyst@ed1f90223c80:~$ pwd
/home/analyst
analyst@ed1f90223c80:~$ ls
logs  projects  reports  temp
analyst@ed1f90223c80:~$ cd /home/analyst/reports
analyst@ed1f90223c80:~/reports$
analyst@ed1f90223c80:~/reports$ ls
users
analyst@ed1f90223c80:~/reports$ cd /home/analyst/reports/users
analyst@ed1f90223c80:~/reports/users$ ls
Q1_added_users.txt  Q1_deleted_users.txt
analyst@ed1f90223c80:~/reports/users$ cat Q1_added_users.txt
employee_id  username  department
1001          bmoreno   Marketing
1026          apatel    Human Resources
1041          cgriffin  Sales
1104          mreed     Information Technology
1177          aezra     Human Resources
1188          noshiro   Finance
analyst@ed1f90223c80:~/reports/users$ cd logs
-bash: cd: logs: No such file or directory
analyst@ed1f90223c80:~/reports/users$ cd /home/analyst/logs
analyst@ed1f90223c80:~/logs$ cat
ls
ls
pwd
pwd
^C
analyst@ed1f90223c80:~/logs$ ls
server_logs.txt
analyst@ed1f90223c80:~/logs$ head server_logs.txt
```

```
analyst@ed1f90223c80:~/logs$ head server_logs.txt
2022-09-28 13:55:55 info      User logged on successfully
2022-09-28 13:56:22 error    The password is incorrect
2022-09-28 13:56:48 warning  The file storage is 75% full
2022-09-28 15:55:55 info      User logged on successfully
2022-09-28 15:56:22 error    The username is incorrect
2022-09-28 15:56:48 warning  The file storage is 90% full
2022-09-28 16:55:55 info      User navigated to settings page
2022-09-28 16:56:22 error    The password is incorrect
2022-09-28 16:56:48 warning  The current user's password expires in 15 days
2022-09-29 13:55:55 info      User logged on successfully
analyst@ed1f90223c80:~/logs$ head server_logs.txt
```